PGCP-AC AUGUST 26
Assignment No- 4

Write Create a Employee ,Department & Project Tables .Add atleast 5 records in each table & Write a following Queries.

```
mysql> create table Employee(emp_id int primary key, emp_name varchar(20), salary int,
location varchar(20));

Query OK, 0 rows affected (0.03 sec)

mysql> insert into Employee(emp_id, emp_name, salary, location)
values(1,'Rahul',75000,'Mumbai'),(2,'Priya',80000,'Pune'),(3,'Virat',70000,'Mumbai'),(4,'Nakul',6
0000,'pune'),(5,'Nikhil',50000,'Mumbai');

Query OK, 5 rows affected (0.01 sec)

Records: 5  Duplicates: 0  Warnings: 0

mysql> create table Department(dept_id int primary key, dept_name varchar(20), emp_id int,
    -> foreign key(emp_id) references Employee(emp_id));

Query OK, 0 rows affected (0.04 sec)

mysql> insert into Department(dept_id, dept_name, emp_id)
values(101,'IT',1),(102,'IT',2),(103,'Comp',3),(104,'Comp',4),(105,'Mech',5);

Query OK, 5 rows affected (0.01 sec)

Records: 5  Duplicates: 0  Warnings: 0

mysql> create table Project(pro_id int primary key, pro_name varchar(30), dept_id int, emp_id
int,
    -> foreign key(emp_id) references Employee(emp_id),
    -> foreign key(dept_id) references Department(dept_id) );

Query OK, 0 rows affected (0.04 sec)
```

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mysql> insert into Project(pro_id, pro_name, dept_id, emp_id) values(10001,'AI Model',101,1),(10002,'ML_&_AI',102,2),(10003,'MYSQL_Model',103,3),(10004,'CYBER_SECURITY',104,4),(10005,'SECURITY',105,5);

Query OK, 5 rows affected (0.01 sec)

Records: 5 Duplicates: 0 Warnings: 0

1. **Write a Relational Algebra expression to perform** Cartesian Product (Cross Join) **between Emp and Dept tables.**

   mysql> select * from employee cross join department;

```
+--------+----------+--------+----------+---------+-----------+--------+
| emp_id | emp_name | salary | location | dept_id | dept_name | emp_id |
+--------+----------+--------+----------+---------+-----------+--------+
|      5 | Nikhil   |  50000 | Mumbai   |     101 | IT        |      1 |
|      4 | Nakul    |  60000 | pune     |     101 | IT        |      1 |
|      3 | Virat    |  70000 | Mumbai   |     101 | IT        |      1 |
|      2 | Priya    |  80000 | Pune     |     101 | IT        |      1 |
|      1 | Rahul    |  75000 | Mumbai   |     101 | IT        |      1 |
|      5 | Nikhil   |  50000 | Mumbai   |     102 | IT        |      2 |
|      4 | Nakul    |  60000 | pune     |     102 | IT        |      2 |
|      3 | Virat    |  70000 | Mumbai   |     102 | IT        |      2 |
|      2 | Priya    |  80000 | Pune     |     102 | IT        |      2 |
|      1 | Rahul    |  75000 | Mumbai   |     102 | IT        |      2 |
|      5 | Nikhil   |  50000 | Mumbai   |     103 | Comp      |      3 |
```

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```
|     4 | Nakul   |  60000 | pune    |    103 | Comp     |     3 |
|     3 | Virat   |  70000 | Mumbai  |    103 | Comp     |     3 |
|     2 | Priya   |  80000 | Pune    |    103 | Comp     |     3 |
|     1 | Rahul   |  75000 | Mumbai  |    103 | Comp     |     3 |
|     5 | Nikhil  |  50000 | Mumbai  |    104 | Comp     |     4 |
|     4 | Nakul   |  60000 | pune    |    104 | Comp     |     4 |
|     3 | Virat   |  70000 | Mumbai  |    104 | Comp     |     4 |
|     2 | Priya   |  80000 | Pune    |    104 | Comp     |     4 |
|     1 | Rahul   |  75000 | Mumbai  |    104 | Comp     |     4 |
|     5 | Nikhil  |  50000 | Mumbai  |    105 | Mech     |     5 |
|     4 | Nakul   |  60000 | pune    |    105 | Mech     |     5 |
|     3 | Virat   |  70000 | Mumbai  |    105 | Mech     |     5 |
|     2 | Priya   |  80000 | Pune    |    105 | Mech     |     5 |
|     1 | Rahul   |  75000 | Mumbai  |    105 | Mech     |     5 |
+-------+---------+--------+---------+--------+----------+-------+
25 rows in set (0.00 sec)
```

2. Write a SQL query to perform an **INNER JOIN** between Emp and Dept tables displaying EmpName and DeptName.

```
mysql> select e.emp_name, d.dept_name from employee as e
    -> inner join department as d
    -> on e.emp_id=d.emp_id;
```

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+----------+-----------+

| emp_name | dept_name |

+----------+-----------+

| Rahul | IT |

| Priya | IT |

| Virat | Comp |

| Nakul | Comp |

| Nikhil | Mech |

+----------+-----------+

5 rows in set (0.00 sec)

3. **Write a SQL query to perform a** LEFT OUTER JOIN **between Emp and Dept tables to display all employees along with their department information, including employees without a department.**

mysql> select e.emp_id, e.emp_name, e.salary, e.location, d.dept_id, d.dept_name

-> from Employee as e

-> left outer join department as d

-> on e.emp_id = d.emp_id;

+--------+----------+--------+----------+---------+-----------+

| emp_id | emp_name | salary | location | dept_id | dept_name |

+--------+----------+--------+----------+---------+-----------+

| 1 | Rahul | 75000 | Mumbai | 101 | IT |

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| 2 | Priya | 80000 | Pune | 102 | IT |

| 3 | Virat | 70000 | Mumbai | 103 | Comp |

| 4 | Nakul | 60000 | pune | 104 | Comp |

| 5 | Nikhil | 50000 | Mumbai | 105 | Mech |

+--------+----------+--------+----------+---------+-----------+

5 rows in set (0.00 sec)

4. **Write a SQL query to perform a** RIGHT OUTER JOIN **between Emp and Dept tables.**

mysql> select e.emp_id, e.emp_name, e.salary, e.location, d.dept_id, d.dept_name

-> from Employee as e

-> right outer join department as d

-> on e.emp_id = d.emp_id;

+--------+----------+--------+----------+---------+-----------+

| emp_id | emp_name | salary | location | dept_id | dept_name |

+--------+----------+--------+----------+---------+-----------+

| 1 | Rahul | 75000 | Mumbai | 101 | IT |

| 2 | Priya | 80000 | Pune | 102 | IT |

| 3 | Virat | 70000 | Mumbai | 103 | Comp |

| 4 | Nakul | 60000 | pune | 104 | Comp |

| 5 | Nikhil | 50000 | Mumbai | 105 | Mech |

+--------+----------+--------+----------+---------+-----------+

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5 rows in set (0.00 sec)

5. Write a SQL query to perform a NATURAL JOIN **between Emp and Dept.**

```
mysql> select * from employee
   -> natural join department;
+--------+----------+--------+----------+---------+-----------+
| emp_id | emp_name | salary | location | dept_id | dept_name |
+--------+----------+--------+----------+---------+-----------+
|      1 | Rahul    |  75000 | Mumbai   |     101 | IT        |
|      2 | Priya    |  80000 | Pune     |     102 | IT        |
|      3 | Virat    |  70000 | Mumbai   |     103 | Comp      |
|      4 | Nakul    |  60000 | pune     |     104 | Comp      |
|      5 | Nikhil   |  50000 | Mumbai   |     105 | Mech      |
+--------+----------+--------+----------+---------+-----------+
5 rows in set (0.00 sec)
```

6. Write a SQL query to perform a CROSS JOIN **between Emp and Project table.**

```
mysql> select *  from employee cross join project;
+--------+----------+--------+----------+--------+---------------+---------+--------+
| emp_id | emp_name | salary | location | pro_id | pro_name      | dept_id | emp_id |
+--------+----------+--------+----------+--------+---------------+---------+--------+
|      5 | Nikhil   |  50000 | Mumbai   |  10001 | AI Model      |     101 |      1 |
```

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| 4 | Nakul | 60000 | pune | 10001 | AI Model | 101 | 1 |

| 3 | Virat | 70000 | Mumbai | 10001 | AI Model | 101 | 1 |

| 2 | Priya | 80000 | Pune | 10001 | AI Model | 101 | 1 |

| 1 | Rahul | 75000 | Mumbai | 10001 | AI Model | 101 | 1 |

| 5 | Nikhil | 50000 | Mumbai | 10002 | ML_&_AI | 102 | 2 |

| 4 | Nakul | 60000 | pune | 10002 | ML_&_AI | 102 | 2 |

| 3 | Virat | 70000 | Mumbai | 10002 | ML_&_AI | 102 | 2 |

| 2 | Priya | 80000 | Pune | 10002 | ML_&_AI | 102 | 2 |

| 1 | Rahul | 75000 | Mumbai | 10002 | ML_&_AI | 102 | 2 |

| 5 | Nikhil | 50000 | Mumbai | 10003 | MYSQL_Model | 103 | 3 |

| 4 | Nakul | 60000 | pune | 10003 | MYSQL_Model | 103 | 3 |

| 3 | Virat | 70000 | Mumbai | 10003 | MYSQL_Model | 103 | 3 |

| 2 | Priya | 80000 | Pune | 10003 | MYSQL_Model | 103 | 3 |

| 1 | Rahul | 75000 | Mumbai | 10003 | MYSQL_Model | 103 | 3 |

| 5 | Nikhil | 50000 | Mumbai | 10004 | CYBER_SECURITY | 104 | 4 |

| 4 | Nakul | 60000 | pune | 10004 | CYBER_SECURITY | 104 | 4 |

| 3 | Virat | 70000 | Mumbai | 10004 | CYBER_SECURITY | 104 | 4 |

| 2 | Priya | 80000 | Pune | 10004 | CYBER_SECURITY | 104 | 4 |

| 1 | Rahul | 75000 | Mumbai | 10004 | CYBER_SECURITY | 104 | 4 |

| 5 | Nikhil | 50000 | Mumbai | 10005 | SECURITY | 105 | 5 |

| 4 | Nakul | 60000 | pune | 10005 | SECURITY | 105 | 5 |

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```
|     3 | Virat    |  70000 | Mumbai   |  10005 | SECURITY      |     105 |     5 |
|     2 | Priya    |  80000 | Pune     |  10005 | SECURITY      |     105 |     5 |
|     1 | Rahul    |  75000 | Mumbai   |  10005 | SECURITY      |     105 |     5 |
+--------+----------+--------+----------+--------+---------------+---------+--------+
```

25 rows in set (0.00 sec)

7. **Write a SQL query to** create a new table EmpBackup with the same structure as Emp but no data**.**

```
mysql> create table EmpBackup like Employee;
Query OK, 0 rows affected (0.02 sec)


mysql> desc empbackup;
+----------+-------------+------+-----+---------+-------+
| Field    | Type        | Null | Key | Default | Extra |
+----------+-------------+------+-----+---------+-------+
| emp_id   | int         | NO   | PRI | NULL    |       |
| emp_name | varchar(20) | YES  |     | NULL    |       |
| salary   | int         | YES  |     | NULL    |       |
| location | varchar(20) | YES  |     | NULL    |       |
+----------+-------------+------+-----+---------+-------+
```

4 rows in set (0.00 sec)

8. **Write a SQL query to copy all data from Emp into EmpBackup table**.

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```
mysql> insert into EmpBackup select * from Employee;

Query OK, 5 rows affected (0.01 sec)

Records: 5  Duplicates: 0  Warnings: 0


mysql> select * from empBackup;

+--------+----------+--------+----------+

| emp_id | emp_name | salary | location |

+--------+----------+--------+----------+

|      1 | Rahul    |  75000 | Mumbai   |

|      2 | Priya    |  80000 | Pune     |

|      3 | Virat    |  70000 | Mumbai   |

|      4 | Nakul    |  60000 | pune     |

|      5 | Nikhil   |  50000 | Mumbai   |

+--------+----------+--------+----------+

5 rows in set (0.00 sec)
```

9. **Write a SQL query to create a new table ProjectArchive with the same structure and data as Project**.

```
mysql> create table ProjectArchive like Project;

Query OK, 0 rows affected (0.04 sec)
```

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```
mysql> insert into ProjectArchive select * from Project;
Query OK, 5 rows affected (0.01 sec)
Records: 5  Duplicates: 0  Warnings: 0

mysql> select * from ProjectArchive;
+--------+---------------+---------+--------+
| pro_id | pro_name      | dept_id | emp_id |
+--------+---------------+---------+--------+
|  10001 | AI Model      |     101 |      1 |
|  10002 | ML_&_AI       |     102 |      2 |
|  10003 | MYSQL_Model   |     103 |      3 |
|  10004 | CYBER_SECURITY |     104 |      4 |
|  10005 | SECURITY      |     105 |      5 |
+--------+---------------+---------+--------+
5 rows in set (0.00 sec)
```

10.Write a SQL query to create an AUTO_INCREMENT sequence for EmpID in the Emp table during table creation.

```
mysql> set foreign_key_checks=0;
Query OK, 0 rows affected (0.00 sec)
mysql> alter table Employee
   -> modify Emp_id int auto_increment;
```

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```
Query OK, 5 rows affected (0.07 sec)

Records: 5  Duplicates: 0  Warnings: 0

mysql> set foreign_key_checks=1;

Query OK, 0 rows affected (0.00 sec)

mysql> select * from employee;

+--------+----------+--------+----------+

| Emp_id | emp_name | salary | location |

+--------+----------+--------+----------+

|      1 | Rahul    |  75000 | Mumbai   |

|      2 | Priya    |  80000 | Pune     |

|      3 | Virat    |  70000 | Mumbai   |

|      4 | Nakul    |  60000 | pune     |

|      5 | Nikhil   |  50000 | Mumbai   |

+--------+----------+--------+----------+

5 rows in set (0.00 sec)

mysql> insert into employee(emp_name,salary,location)
values('Yash',70000,'Pune');

Query OK, 1 row affected (0.01 sec)

mysql> select * from employee;

+--------+----------+--------+----------+

| Emp_id | emp_name | salary | location |
```

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```
+--------+----------+--------+----------+
|      1 | Rahul    |  75000 | Mumbai   |
|      2 | Priya    |  80000 | Pune     |
|      3 | Virat    |  70000 | Mumbai   |
|      4 | Nakul    |  60000 | pune     |
|      5 | Nikhil   |  50000 | Mumbai   |
|      6 | Yash     |  70000 | Pune     |
+--------+----------+--------+----------+
6 rows in set (0.00 sec)
```

11. Write a SQL query using a subquery **to find all employees whose salary is greater than the** average salary of all employees**.**

```
mysql> select Emp_id, Emp_name, Salary, Location
   -> from employee
   -> where salary > (select avg(salary) from employee);
+--------+----------+--------+----------+
| Emp_id | Emp_name | Salary | Location |
+--------+----------+--------+----------+
|      1 | Rahul    |  75000 | Mumbai   |
|      2 | Priya    |  80000 | Pune     |
|      3 | Virat    |  70000 | Mumbai   |
|      6 | Yash     |  70000 | Pune     |
```

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+--------+----------+--------+----------+

4 rows in set (0.00 sec)

12.Write a SQL query using a subquery in WHERE clause **to find employees working in departments located in 'Mumbai'.**

```
mysql> select emp_name, location from employee
   -> where emp_id in(select emp_id from employee where location='Mumbai');
+----------+----------+
| emp_name | location |
+----------+----------+
| Rahul    | Mumbai   |
| Virat    | Mumbai   |
| Nikhil   | Mumbai   |
+----------+----------+
3 rows in set (0.00 sec)
```

13.Write a SQL query to display all departments where the number of employees is greater than 5**, using a subquery.**

```
mysql> select dept_name, count(emp_id) as "Total Employees"
   -> from Department group by  dept_name;
+-----------+-----------------+
| dept_name | Total Employees |
+-----------+-----------------+
| IT        |               2 |
```

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| Comp | 2 |

| Mech | 1 |

+----------+----------------+

3 rows in set (0.00 sec)

14. Write a subqueries for find out Max salary of employee.

```
mysql> select Emp_id, Emp_name, Salary, Location from Employee
   -> where salary = (select max(salary) from employee);
```

+--------+----------+--------+----------+

| Emp_id | Emp_name | Salary | Location |

+--------+----------+--------+----------+

| 2 | Priya | 80000 | Pune |

+--------+----------+--------+----------+

1 row in set (0.00 sec)

15.Write a self join query for each table.

```
mysql> select e.emp_name as Employee_name, d.emp_name as
Employee_name2, e.location from Employee as e
   -> left join Employee as d
   -> on e.location=d.location and e.emp_id < d.emp_id;
```

+--------------+---------------+----------+

| Employee_name | Employee_name2 | location |

+--------------+---------------+----------+

| Rahul | Virat | Mumbai |

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| Rahul | Nikhil | Mumbai |

| Priya | Nakul | Pune |

| Priya | Yash | Pune |

| Virat | Nikhil | Mumbai |

| Nakul | Yash | pune |

| Nikhil | NULL | Mumbai |

| Yash | NULL | Pune |

+---------------+----------------+----------+

8 rows in set (0.00 sec)

mysql> select d.dept_name as Dept_A, c.dept_name as Dept_B from Department as d

-> Inner join department as c

-> on d.dept_name < c.dept_name;

+--------+--------+

| Dept_A | Dept_B |

+--------+--------+

| Comp | IT |

| Comp | IT |

| Comp | IT |

| Comp | IT |

| Comp | Mech |

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```
| Comp   | Mech   |
| IT     | Mech   |
| IT     | Mech   |
+--------+--------+
8 rows in set (0.00 sec)
```

```
mysql> select p.pro_name as Project_A, q.pro_name as Project_B from Project as
p
   -> Inner join project as q
   -> on p.pro_name < q.pro_name;
+---------------+---------------+
| Project_A     | Project_B     |
+---------------+---------------+
| CYBER_SECURITY | ML_&_AI       |
| AI Model      | ML_&_AI       |
| CYBER_SECURITY | MYSQL_Model   |
| ML_&_AI       | MYSQL_Model   |
| AI Model      | MYSQL_Model   |
| AI Model      | CYBER_SECURITY |
| CYBER_SECURITY | SECURITY      |
| MYSQL_Model   | SECURITY      |
| ML_&_AI       | SECURITY      |
```

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| AI Model | SECURITY |

+----------+----------+

10 rows in set (0.00 sec)

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